

LLMs for Economics Research

A very opinionated tour

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Applied Reading Group
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Why am I giving a talk about AI?

- I am not an AI expert.
- I am an *infovore*: I spend too much time online.
- Here are some things I have found useful.

A great course to get started

10-202: Introduction to Modern AI

by Zico Kolter

► modernaicourse.org

Course Contents

- AI history, linear algebra and PyTorch
- Supervised learning and neural networks
- Language models and Transformers
- Post-training and reasoning
- AI safety, security and the future of AGI

1950s–1960s: Rosenblatt's perceptron



“the machine would be the first device to think as the human brain. As do human beings, Perceptron will make mistakes at first, but will grow wiser as it gains experience”

The fracturing of AI: 80s–00s

- 1980s — neural networks trained via backpropagation, convolutional and recurrent networks
- 1997 — Deep Blue beat Garry Kasparov
- 2000s — Ads / recommendations
- 2005/2007 — DARPA Grand Challenges (autonomous driving)

The rebirth of AI: 10s–present

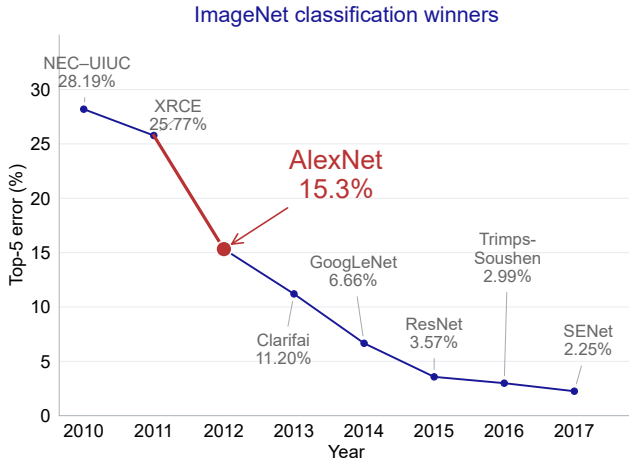
- 2012 — AlexNet (emergence of deep learning)
- 2014 — Seq2Seq (machine translation)
- 2016 — AlphaGo
- 2017 — Transformer paper
- 2022 — ChatGPT

2012: AlexNet



Fei-Fei Li

► TED talk



2017: The Transformer

Attention Is All You Need

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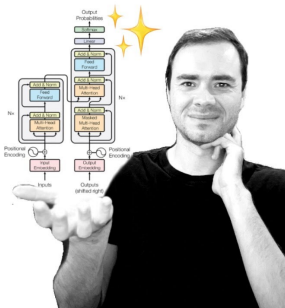
► [Original paper](#)

► [The Annotated Transformer \(PyTorch\)](#)

The architecture behind the **T** in GPT.

A great way to understand GPT

LET'S BUILD GPT.
FROM SCRATCH.
IN CODE.
SPELLED OUT.



A very clear, step-by-step coding course (about 2 hours).

► Lecture: Let's build GPT



Deep learning for solving economic models

Deep Learning for Solving Economic Models

Jesús Fernández- Villaverde

JOURNAL OF ECONOMIC LITERATURE
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Abstract

The ongoing revolution in deep learning is reshaping research across many fields, including economics. Its effects are especially clear in solving dynamic economic models. These models often lack closed-form solutions, so economists have long relied on numerical methods such as value function iteration, perturbation, and projection techniques. Unfortunately, these approaches suffer from the curse of dimensionality, which makes global solutions computationally infeasible as the number of state variables increases. Deep learning offers a different approach: flexible tools that solve dynamic economic models by minimizing residuals in equilibrium conditions and that can handle high-dimensional problems. This development promises to broaden the scope of quantitative economics. I illustrate the approach using the neoclassical growth model.

The problem

- Let us suppose we want to approximate (“learn”) an unknown function:

$$y = f(\mathbf{x})$$

where y is a scalar and $\mathbf{x} = \{x_0 = 1, x_1, x_2, \dots, x_N\}$ a vector (why a constant?).

- We care about the case when N is large (possibly in the thousands!).
- Easy to extend to the case where y is a vector (e.g., a probability distribution), but notation becomes cumbersome.
- In economics, $f(\mathbf{x})$ can be a value function, a policy function, a pricing kernel, a conditional expectation, a classifier, ...

A neural network

- A neural network is an approximation to $f(\mathbf{x})$ of the form:

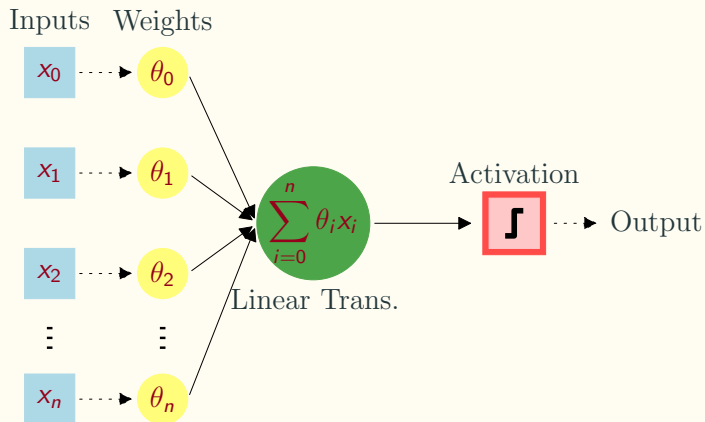
$$y \cong g^{NN}(\mathbf{x}; \theta) = \theta_0 + \sum_{m=1}^M \theta_m \phi(z_m)$$

where $\phi(\cdot)$ is an activation function and:

$$z_m = \sum_{n=0}^N \theta_{n,m} x_n$$

- The x_n 's are known as the features of the data, which belong to a feature space $\mathcal{X}$.
- The $\phi(z_m)$'s are known as the representation of the data.
- M is known as the width of the model (wide vs. thin networks).
- “Training” the network: selecting θ such that $g^{NN}(\mathbf{x}; \theta)$ is as close to $f(\mathbf{x})$ as possible given some relevant metric (e.g., the ℓ_2 norm).

Flow representation



Deep learning I

- A deep learning network is a *multilayer* composition of $J > 1$ neural networks:

$$z_m^0 = \theta_{0,m}^0 + \sum_{n=1}^N \theta_{n,m}^0 x_n$$

and

$$z_m^1 = \theta_{0,m}^1 + \sum_{m=1}^{M^{(1)}} \theta_m^1 \phi^1(z_m^0)$$

...

$$y \cong g^{DL}(\mathbf{x}; \theta) = \theta_0^J + \sum_{m=1}^{M^{(J)}} \theta_m^J \phi^J(z_m^{J-1})$$

where the $M^{(1)}, M^{(2)}, \dots$ and $\phi^1(\cdot), \phi^2(\cdot), \dots$ are possibly different across each layer of the network.

From a Few Parameters to Many Billions

- Griliches's Law: "More than five variables in a cross section yields garbage."
- Von Neumann: "With four parameters I can fit an elephant, and with five I can make him wiggle his trunk."
- GPT-3 (2020): 175 billion parameters; Llama 3.1 (2024): 405 billion.
- Qwen3.5 (2026): 397 billion parameters; 17 billion active per token.
- Kimi K2.5 (2026): 1,000 billion parameters; 32 billion active per token.

Overparametrization: But is this not a problem?

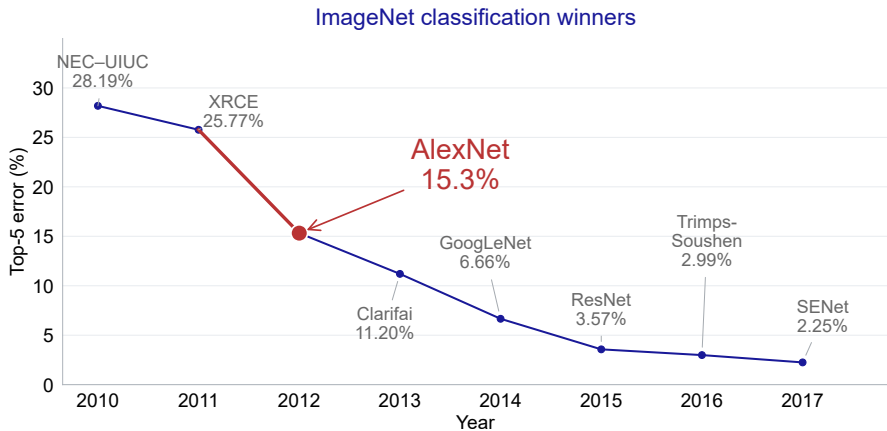
The Blessings of Overparameterization: Applications in Solving Economic Models*

Mahdi Ebrahimi Kahou[†]

Jesús Fernández-Villaverde[‡]

- Think of *costates*: we return to the original economic variables.
- Interpret and check the resulting policy or value function.

It just works!!!



An LLM predicts the next token

Context $\longrightarrow$ next token $\longrightarrow$ repeat

$$P_{\theta}(t_{n+1} \mid t_1, \dots, t_n)$$

- Text and code: words, word parts, punctuation.
- Embeddings represent tokens numerically.
- Multimodal models also use image and audio units.

Give the model tools

LLM + skills + tools + feedback

Decide $\longrightarrow$ act $\longrightarrow$ observe $\longrightarrow$ repeat

- Skills: reusable instructions.
- Tools: search, read files, run code.
- Feedback: check the result and try again.

Context: correct, complete and concise

$$\text{Performance} \sim \frac{\text{correctness}^2 \times \text{completeness}}{\text{size}}$$

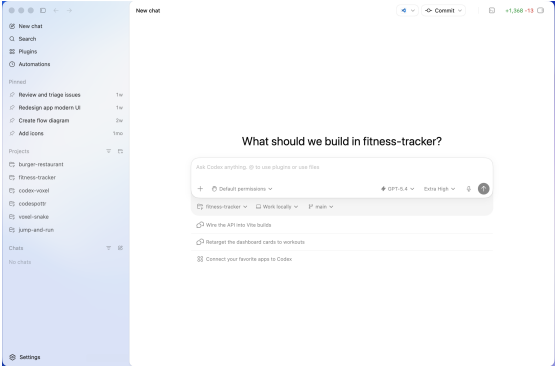
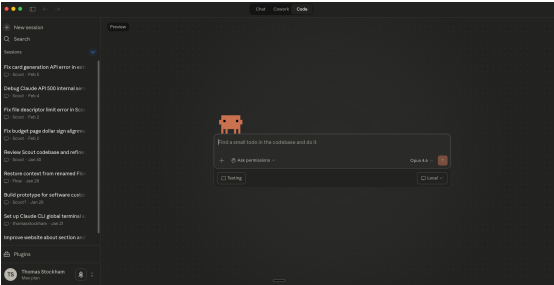
A rule of thumb, not an estimated law.

- Use reliable facts and current instructions.
- Include what the task needs.
- Long context can bury useful information. Keep a short record of decisions.

How I use these tools

- I am not very good at prompting. I just iterate a lot!
- A few habits and tools help.

Claude Code and Codex



Pedro H. C. Sant'Anna

My Claude Code Setup

Pedro H. C. Sant'Anna · Emory University

Actively developed and maintained. This summarizes how I use Claude Code for academic work — slides, papers, data analysis, and more. Adopted by research groups across economics, energy, political science, and engineering — 15+ research groups at the March 2026 survey; the repository has since passed 2,900 forks and 1,500 stars (GitHub, 2026-08-24). A fork is not a research group — those two figures measure different things. I keep learning and updating these files.

A foundation for AI-assisted academic work using [Claude Code](#). You describe what you want — lecture slides, a research paper, a data analysis, a replication package — and Claude plans the approach, runs specialized agents, fixes issues, verifies quality, and presents results — goal-first and gate-enforced. Like a contractor who handles the entire job, with you as the auditor of the disagreements the review loop surfaces. It is deliberately not an autonomous daemon: every loop is started by you or a skill, never on its own.

[Read the full guide](#)

► [My Claude Code Setup](#)

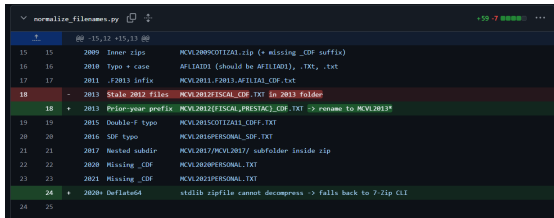
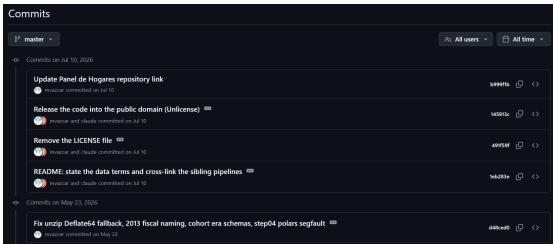
Claude Blattman

Tools I use to manage my schedule, communication, and complex projects with large teams across several countries. Examples and tutorials for productivity and sanity.

By [Chris Blattman](#), political economist at UChicago Harris. In January 2026 I started building AI workflows with Claude Code. What seemed impossible then — inbox triage, meeting capture, proposal drafting, project dashboards, trip planning — is running today. Claude Blattman isn't real, but the tools are. I've never coded in my life, so if I can do this you can.

Adopt modern workflows: Git and GitHub

- Get a GitHub account. Use Git to track and review changes.



Some skills I find really useful

- I have ADHD: one step at a time.
- GOV.UK text: plain English.
- Scott Cunningham's Mixtape tools: Referee 2.
- No-mistakes: check code before delivery.

The value of good questions

- I thought I had more good questions than I actually do.
- Simple models now cost me almost nothing to implement.
- Good questions are still hard to find.

Are ideas getting harder to find?

Are Ideas Getting Harder to Find?[†]

By NICHOLAS BLOOM, CHARLES I. JONES, JOHN VAN REENEN,
AND MICHAEL WEBB*

This paper applies the growth accounting of Solow (1957) to the production function for new ideas. The basic insight can be explained with a simple equation, highlighting a stylized view of economic growth that emerges from idea-based growth models:

$$\begin{array}{ccccc} \text{Economic growth} & = & \text{Research productivity} & \times & \text{Number of researchers.} \\ \text{e.g., 2\% or 5\%} & & \downarrow (\text{falling}) & & \uparrow (\text{rising}) \end{array}$$

Weak links: verification is the bottleneck

Past Automation and Future A.I.: How Weak Links Tame the Growth Explosion

Charles I. Jones

Stanford GSB and NBER

Christopher Tonetti

Stanford GSB and NBER

- In knowledge production, the weak link is now verification, not implementation.

Verification is the bottleneck

'coarse!

Free software; model usage costs apply.

r refine

Paid reviews; free trial available.

Verification: data, code and journal standards



► Journals endorsing the standard

1. American Economic Journal: Applied Economics  
2. American Economic Journal: Economic Policy  
3. American Economic Journal: Macroeconomics  
4. American Economic Journal: Microeconomics  
5. American Economic Review  
6. American Economic Review: Insights  
7. Canadian Journal of Economics 
8. Econometrica 
9. Econometrics Journal
10. Economic Inquiry 
11. Economic Journal 
12. Journal of Economic Literature  
13. Journal of Economic Perspectives  
14. Journal of Political Economy 
15. Journal of the European Economic Association 
16. Quantitative Economics 
17. Review of Economic Studies 
18. Review of Financial Studies 
19. Theoretical Economics 

Don't be a meat proxy

Niklas Gruhn



Don't be a meat proxy

Aug 03, 2026

Too often I ask a question in Slack or leave feedback under a merge/pull request or argue with friends in a WhatsApp group and get back:

| *Claude said: [giant response verbatim]*

Please don't do this. I mean, I've done this. But I've been on the receiving end too many times now. This is not adding value. I can talk to Claude myself. It's going to be faster and I get to control the context. I don't need a meat proxy in between.